

Recursive Kalman-Based Temporal Adaptation on Rotating 3D Two-Moons Domains

This project evaluates whether evolving neural decision boundaries can be modeled as latent temporal trajectories using anchor-point observations and recursive Kalman filtering. The benchmark extends the classic Two Moons dataset from 2D to 3D and introduces sinusoidal rotational domain drift over time.

Overview

1. Generate temporally drifting 3D Two-Moons domains
2. Train one neural classifier per domain
3. Evaluate each classifier on fixed 3D anchor probes
4. Treat anchor-response vectors as temporal observations
5. Fit a Kalman filter to the evolving anchor trajectories
6. Predict the next domain's latent decision-boundary state
7. Recursively adapt the classifier using predicted future anchor responses

Synthetic Drift Environment

The dataset consists of: 3D embedded Two Moons manifolds Sinusoidal rotational drift around the Z-axis
Domain-specific stochastic perturbations The latent rotation evolves according to:

$$\theta(t) = \pi \sin(2\pi t / T_{\text{total}}) + \epsilon_t$$

Anchor Representation

Each classifier is evaluated on a fixed set of anchor probes distributed throughout the 3D input space. For each anchor, the classifier outputs $P(y=1|x)$. The resulting probability vector becomes the temporal observation used for Kalman filtering.

Recursive Closed-Loop Adaptation

The framework operates recursively: Past observed domains estimate latent temporal dynamics Kalman filtering predicts future anchor-response states The classifier is adapted toward the predicted future boundary Newly observed domains are incorporated back into the temporal history

Dependencies

```
pip install numpy matplotlib torch scikit-learn pykalman
```

Running

```
python kalman_3d_recursive.py
```

Main Files

File	Purpose
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Key Research Question

Can evolving neural decision boundaries be represented as latent temporal trajectories and recursively generalized to future domains using anchor-based Kalman filtering?

Notes

The framework models temporal dynamics in anchor-response space rather than forecasting neural weights directly. Anchor trajectories can exhibit nonlinear step-like transitions due to moving decision boundaries. Recursive Kalman adaptation stabilizes future-domain performance under structured rotational drift.